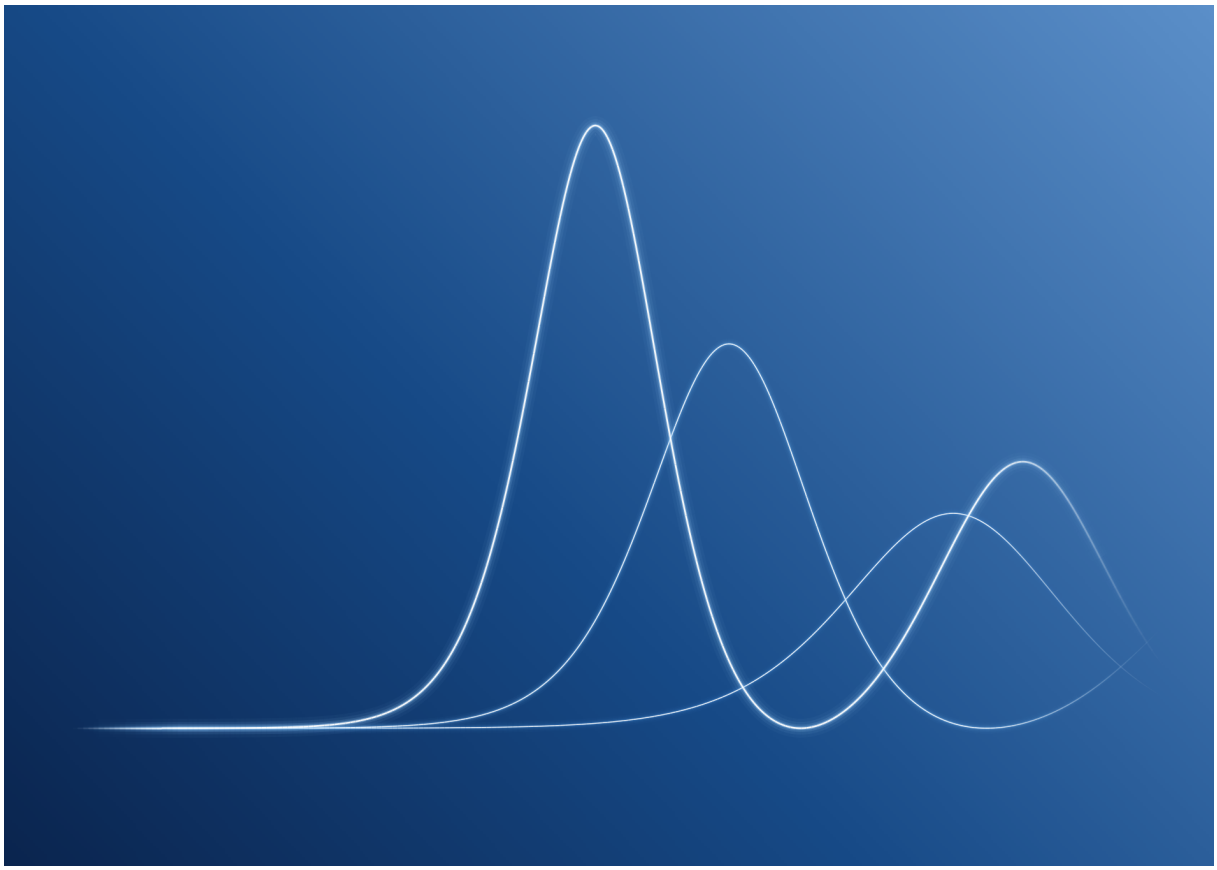


# Maxwell Bloch Studio

Equations Reference

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This document derives the dimensionless Maxwell–Bloch equations used by the solver, starting from the general two-level Hamiltonian and proceeding through the rotating-wave and slowly-varying-envelope approximations to the final dimensionless system.

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## 1 Hamiltonian

Let $|2\rangle$ denote the upper state and $|1\rangle$ the lower state of a two-level system with transition frequency $\omega_0$. The free Hamiltonian is

$$\hat{H}_0 = \frac{\hbar\omega_0}{2}\hat{\sigma}_z, \qquad \hat{\sigma}_z = |2\rangle\langle 2| - |1\rangle\langle 1|, \tag{1}$$

with raising and lowering operators

$$\hat{\sigma}_+ = |2\rangle\langle 1|, \qquad \hat{\sigma}_- = |1\rangle\langle 2|. \tag{2}$$

The electric dipole operator is written in terms of the transition matrix elements $\boldsymbol{d}_{21} = \langle 2|\hat{\boldsymbol{d}}|1\rangle$ and $\boldsymbol{d}_{12} = \boldsymbol{d}_{21}^*$ as

$$\hat{\boldsymbol{d}} = \boldsymbol{d}_{21}\,\hat{\sigma}_+ + \boldsymbol{d}_{12}\,\hat{\sigma}_-. \tag{3}$$

Writing the electric field along polarization unit vector $\hat{\boldsymbol{e}}$ as $\boldsymbol{E} = E\hat{\boldsymbol{e}}$ and defining the scalar transition matrix element

$$d = \boldsymbol{d}_{21}\cdot\hat{\boldsymbol{e}}, \tag{4}$$

the interaction Hamiltonian takes the form

$$\hat{H}_{\text{int}} = -\hat{\boldsymbol{d}}\cdot\boldsymbol{E} = -\big(d\,\hat{\sigma}_+ + d^*\hat{\sigma}_-\big)E. \tag{5}$$

The total Hamiltonian is $\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}$.

## 2 Density-Matrix Equations of Motion

The state of the medium is described by the density operator

$$\hat{\rho} = \sum_{m,n\,\in\,\{1,2\}} \rho_{mn}\,|m\rangle\langle n|, \tag{6}$$

with population inversion

$$w = \rho_{22} - \rho_{11}, \qquad \rho_{21} = \rho_{12}^*. \tag{7}$$

From the Liouville–von Neumann equation $i\hbar\dot{\hat{\rho}} = [\hat{H},\hat{\rho}]$, the matrix element $\langle 1|\,[\hat{H},\hat{\rho}]\,|2\rangle$ gives

$$\begin{aligned} i\hbar\dot{\rho}_{12} &= \langle 1|[\hat{H},\hat{\rho}]|2\rangle \\ &= -\hbar\omega_0\rho_{12} - d^*E\,(\rho_{22}-\rho_{11}), \end{aligned} \tag{8}$$

giving

$$\dot{\rho}_{12} = i\omega_0\rho_{12} + \frac{id^*}{\hbar}\,E\,w. \tag{9}$$

Since $\rho_{21} = \rho_{12}^*$, the complex conjugate gives immediately

$$\dot{\rho}_{21} = -i\omega_0\rho_{21} - \frac{id}{\hbar}\,E\,w. \tag{10}$$

For the inversion, taking $\langle 2|[\hat{H},\hat{\rho}]|2\rangle - \langle 1|[\hat{H},\hat{\rho}]|1\rangle$:

$$i\hbar\dot{w} = 2\big(d^*E\,\rho_{21} - dE\,\rho_{12}\big), \tag{11}$$

so that

$$\dot{w} = \frac{2i}{\hbar}\big(dE\,\rho_{12} - d^*E\,\rho_{21}\big). \tag{12}$$

# 3 Maxwell Wave Equations

## 3.1 Maxwell's equations

In a medium with no free charges or currents, Maxwell's equations are

$$\nabla \cdot \mathbf{D} = 0, \tag{13}$$
$$\nabla \cdot \mathbf{B} = 0, \tag{14}$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{15}$$
$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}, \tag{16}$$

with constitutive relations

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \qquad \mathbf{B} = \mu_0 \big(\mathbf{H} + \mathbf{M}\big). \tag{17}$$

## 3.2 General wave equations

Taking the curl of the curl equations and using the constitutive relations gives the general wave equations for $\mathbf{E}$ and $\mathbf{B}$:

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial^2 \mathbf{P}}{\partial t^2} + \mu_0 \nabla \times \frac{\partial \mathbf{M}}{\partial t}, \tag{18}$$

$$\nabla^2 \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = -\mu_0 \nabla \times \left( \frac{\partial \mathbf{P}}{\partial t} + \nabla \times \mathbf{M} \right). \tag{19}$$

## 3.3 Reduction to one dimension

We consider a plane wave propagating along $z$ with fields transverse to $z$. For the **electric-dipole** case the medium has no magnetization ($\mathbf{M} = 0$), so the curl term in Eq. (18) vanishes and the wave equation reduces to

$$\frac{\partial^2 E}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} = \mu_0 \frac{\partial^2 P}{\partial t^2}. \tag{20}$$

The macroscopic polarization is related to the density operator through

$$P = n_0 \operatorname{Tr}(\hat{\rho}\, \hat{\boldsymbol{d}} \cdot \hat{\boldsymbol{e}}) = n_0 \big(\rho_{12}\, d + \rho_{21}\, d^*\big). \tag{21}$$

# 4 Slowly Varying Envelope and Rotating-Wave Approximations

We apply two standard approximations: the **slowly varying envelope approximation** (SVEA) to the field, and the **rotating-wave approximation** (RWA) to the atom–field coupling.

### 4.1 Envelope decomposition

The field and coherence $\rho_{21}$ are written as slowly varying envelopes modulated by the carrier $e^{i(kz-\omega t)}$:

$$E(z,t) = \tfrac{1}{2}A(z,t)\,e^{i(kz-\omega t)} + \text{c.c.}, \tag{22}$$

$$\rho_{21}(z,t) = \tfrac{1}{2}R(z,t)\,e^{i(kz-\omega t)}, \tag{23}$$

where $k = \omega/c$ is the carrier wave number, and $A$ and $R$ satisfy

$$\left|\frac{\partial A}{\partial t}\right| \ll \omega|A|, \quad \left|\frac{\partial A}{\partial z}\right| \ll k|A|, \tag{24}$$

and similarly for $R$. The factor of $\frac{1}{2}$ ensures that $A$ equals the peak amplitude of the field.

### 4.2 Equations of motion

Substituting the envelope decomposition into Eq. (10), collecting terms at the resonant frequency $e^{i(kz-\omega t)}$, and dropping the fast counter-rotating terms (RWA). we define the detuning

$$\Delta = \omega - \omega_0, \tag{25}$$

the coherence equation becomes

$$\dot{R} = i\Delta R - \frac{id}{\hbar}\,A\,w. \tag{26}$$

Substituting into Eq. (12) and keeping only the two resonant products (one from each factor of the field and coherence):

$$\dot{w} = \frac{i}{2\hbar}\big(dAR^* - d^*A^*R\big). \tag{27}$$

Substituting into the wave equation (20), dropping second-order derivatives of $A$ (SVEA), and using $k = \omega/c$:

$$\frac{\partial A}{\partial z} + \frac{1}{c}\frac{\partial A}{\partial t} = \frac{i\omega\, n_0 d^*}{2\epsilon_0 c}\,R. \tag{28}$$

### 4.3 Relaxation and pumping

In practice the two-level medium is coupled to its environment, leading to population relaxation on a timescale $T_1$ and dephasing on a timescale $T_2$. These are added phenomenologically. A pump term $\Lambda_n$ drives the inversion away from its equilibrium value $w_0$, and the coherence relaxes toward its equilibrium value $R_0$. We also set the detuning $\Delta = 0$, corresponding to exact resonance between the carrier and the atomic transition. The full Maxwell–Bloch system is:

$$\frac{\partial w}{\partial t} = \frac{i}{2\hbar}\big(dAR^* - d^*A^*R\big) - \frac{w - w_0}{T_1} + \Lambda_n, \tag{29}$$

$$\frac{\partial R}{\partial t} = -\frac{id}{\hbar}\,A\,w - \frac{R - R_0}{T_2}, \tag{30}$$

$$\frac{\partial A}{\partial z} + \frac{1}{c}\frac{\partial A}{\partial t} = \frac{i\omega\, n_0 d^*}{2\epsilon_0 c}\,R. \tag{31}$$

# 5 Retarded-Time Coordinates

We move to a frame co-propagating with the field by introducing the retarded time $\tau = t - z/c$, keeping $z$ as the propagation coordinate. Under this transformation

$$\left.\frac{\partial}{\partial z}\right|_t = \left.\frac{\partial}{\partial z}\right|_\tau - \frac{1}{c}\left.\frac{\partial}{\partial \tau}\right|_z, \qquad \left.\frac{\partial}{\partial t}\right|_z = \left.\frac{\partial}{\partial \tau}\right|_z, \tag{32}$$

so the operator $\partial_z|_t + c^{-1}\partial_t|_z$ reduces to $\partial_z|_\tau$, and the field equation becomes a pure spatial propagation equation. The full Maxwell–Bloch system in retarded-time coordinates is:

$$\frac{\partial w}{\partial \tau} = \frac{i}{2\hbar}\left(dAR^* - d^*A^*R\right) - \frac{w - w_0}{T_1} + \Lambda_n, \tag{33}$$

$$\frac{\partial R}{\partial \tau} = -\frac{id}{\hbar}\,A\,w - \frac{R - R_0}{T_2}, \tag{34}$$

$$\frac{\partial A}{\partial z} = \frac{i\omega\, n_0 d^*}{2\epsilon_0 c}\,R. \tag{35}$$

# 6 Dimensionless Form

## 6.1 Scaled variables

The inversion $w$ and coherence $R$ are already dimensionless. We introduce dimensionless forms of the remaining variables by normalising lengths to the sample length $L$, times to the characteristic scale $T_0$, and the field amplitude to $\hbar/dT_0$:

$$z' = \frac{z}{L}, \qquad \tau' = \frac{\tau}{T_0}, \qquad T_1' = \frac{T_1}{T_0}, \qquad T_2' = \frac{T_2}{T_0}, \qquad \Lambda_n' = T_0\Lambda_n. \tag{36}$$

To absorb the imaginary unit in the field equation we define the dimensionless field amplitude

$$A' = \frac{A}{A_0}, \tag{37}$$

where the field amplitude scale is

$$A_0 = -i\frac{\hbar}{d\,T_0}. \tag{38}$$

## 6.2 Cooperation time $T_R$

Substituting Eqs. (37)–(38) into Eq. (35):

$$\frac{\partial A'}{\partial z'} = -\eta\,R, \tag{39}$$

where

$$\eta = \frac{T_0}{T_R}, \tag{40}$$

and the **cooperation time** is defined as

$$\boxed{T_R = \frac{2\epsilon_0 c\hbar}{\omega\,|d|^2\,n_0\,L}.} \tag{41}$$

### 6.3 Final dimensionless equations

Substituting Eq. (37) into Eqs. (33)–(35) and expressing all times in units of $T_0$:

$$\frac{\partial w}{\partial \tau'} = \mathrm{Re}(A' R^*) - \frac{w - w_0}{T_1'} + \Lambda_n', \tag{42}$$

$$\frac{\partial R}{\partial \tau'} = -A'\, w - \frac{R - R_0}{T_2'}, \tag{43}$$

$$\frac{\partial A'}{\partial z'} = -\eta R. \tag{44}$$

**Note**

No explicit $\hbar$, $d$, or $T_0$ appear in Eqs. (42)–(44). All microscopic parameters are absorbed into $\eta$ and the dimensionless field amplitude $A'$.

### 6.4 Symbol table

| Symbol | Meaning | Range | Units |
|---|---|---|---|
| $A_0 = -i\hbar/dT_0$ | Field amplitude scale | $\mathbb{C}$ | (V/m) |
| $A' = A/A_0$ | Dimensionless field envelope | $\mathbb{C}$ | — |
| $R(z', \tau')$ | Coherence envelope ($\rho_{21} = \frac{1}{2} R\, e^{i(kz-\omega t)}$) | $\mathbb{C}$ | — |
| $w(z', \tau')$ | Population inversion $\rho_{22} - \rho_{11}$ | $\mathbb{R}$ | — |
| $z' = z/L$ | Normalised propagation distance | [0, 1] | — |
| $\tau' = \tau/T_0$ | Normalised retarded time | $\mathbb{R}^+$ | — |
| $T_0$ | Physical reference time scale | $\mathbb{R}^+$ | (s) |
| $T_1' = T_1/T_0$ | Longitudinal relaxation time (may vary with $\tau'$) | $\mathbb{R}^+$ | — |
| $T_2' = T_2/T_0$ | Transverse dephasing time (may vary with $\tau'$) | $\mathbb{R}^+$ | — |
| $T_R$ | Cooperation time (Eq. 41) | $\mathbb{R}^+$ | (s) |
| $\eta = T_0/T_R$ | Dimensionless coupling parameter | $\mathbb{R}^+$ | — |
| $\Lambda_n' = T_0 \Lambda_n$ | Dimensionless pump rate | $\mathbb{R}$ | — |
| $w_0$ | Equilibrium inversion | $\mathbb{R}$ | — |

## 7 Initial and Boundary Conditions

### 7.1 General case

The Maxwell–Bloch system (42)–(44) requires:

- an **initial condition** for $R$ at $\tau' = 0$ for all $z'$,
- a **boundary condition** for $A'$ at $z' = 0$ for all $\tau'$.

The inversion $w$ is always real since $\rho_{11}$ and $\rho_{22}$ are real probabilities. In the general case $R$ and $A'$ are complex.

### 7.2 Real-variable reduction

We now show that under a natural choice of initial and boundary conditions, $R$ and $A'$ remain real for all $\tau'$ and $z'$, leading to a significant simplification.

Suppose the initial condition for the coherence satisfies

$$R(z', 0) \in \mathbb{R}, \tag{45}$$

and the boundary field is real:

$$A'(0, \tau') \in \mathbb{R}. \tag{46}$$

Note that $A'$ real corresponds to $A$ purely imaginary, since $A' = A/A_0$ with $A_0 = -i\hbar/dT_0$.

Inspecting the equations: if $w$, $R$, and $A'$ are all real at some $\tau'$, then the right-hand sides of Eqs. (42)–(44) are all real, so the variables remain real for all subsequent times and positions. Under these conditions $\mathrm{Re}(A'R^*) = A'R$, and the Maxwell–Bloch system reduces to:

$$\frac{\partial w}{\partial \tau'} = A'R - \frac{w - w_0}{T_1'} + \Lambda_n', \tag{47}$$

$$\frac{\partial R}{\partial \tau'} = -A'\, w - \frac{R - R_0}{T_2'}, \tag{48}$$

$$\frac{\partial A'}{\partial z'} = -\eta R. \tag{49}$$

**Note**

Equations (47)–(49) are entirely real and require no complex arithmetic. This is the system solved by the application. The equations hold unchanged when the relaxation times depend on time, $T_1' = T_1'(\tau')$ and $T_2' = T_2'(\tau')$. How the boundary field, initial conditions, pump profile, and relaxation times are entered is described in the User Guide.

## 8 Recovering the Physical Fields

We define the field and polarization amplitude scales

$$E_0 = \frac{\hbar}{|d|\, T_0}, \qquad P_0 = n_0\, |d|. \tag{50}$$

Given the dimensionless solution $A'(z', \tau')$, the instantaneous electric field and macroscopic polarization are recovered as

$$E(z, t) = E_0\, \mathrm{Re}\big[-iA'(z', \tau')\, e^{i(kz - \omega t)}\big]\,, \tag{51}$$

$$P(z, t) = P_0\, \mathrm{Re}\big[R(z', \tau')\, e^{i(kz - \omega t)}\big]\,. \tag{52}$$

Since the carrier $e^{i(kz-\omega t)}$ oscillates much faster than the envelopes $A'$ and $R$, physical detectors respond only to the slowly varying amplitudes. The observable intensity is therefore proportional to $|A'|^2$.

In the real case (Subsection 7.2), and taking $d$ real, $A'$ and $R$ are real, and the fields reduce to

$$E(z,t) = E_0\, A'(z',\tau')\sin(kz-\omega t), \tag{53}$$

$$P(z,t) = P_0\, R(z',\tau')\cos(kz-\omega t). \tag{54}$$

## 9 Magnetic-Dipole Case

All preceding sections were derived for the electric-dipole case. The magnetic-dipole (maser) case follows from the substitutions

$$E \;\to\; cB, \qquad d \;\to\; \frac{m}{c}, \qquad P \;\to\; \frac{M}{c}, \tag{55}$$

where $m = \boldsymbol{m}_{21}\cdot\hat{\boldsymbol{b}}$ is the magnetic transition dipole matrix element and $\hat{\boldsymbol{b}}$ is the magnetic field polarization unit vector. Under these replacements the coupling $dE = (m/c)(cB) = mB$ is preserved, and the dimensionless equations (47)–(49) hold unchanged in form.

The table below summarises the correspondence between the two cases at each stage of the derivation.

| Quantity | Electric | Magnetic |
|---|---|---|
| *Microscopic quantities* | | |
| Interaction Hamiltonian | $\hat{H}_{\text{int}} = -\hat{\boldsymbol{d}}\cdot\boldsymbol{E}$ | $\hat{H}_{\text{int}} = -\hat{\boldsymbol{m}}\cdot\boldsymbol{B}$ |
| Coupling constant | $d = \boldsymbol{d}_{21}\cdot\hat{\boldsymbol{e}}$ | $m = \boldsymbol{m}_{21}\cdot\hat{\boldsymbol{b}}$ |
| 1D wave equation | $\dfrac{\partial^2 E}{\partial z^2} - \dfrac{1}{c^2}\dfrac{\partial^2 E}{\partial t^2} = \mu_0\dfrac{\partial^2 P}{\partial t^2}$ | $\dfrac{\partial^2 B}{\partial z^2} - \dfrac{1}{c^2}\dfrac{\partial^2 B}{\partial t^2} = \mu_0\dfrac{\partial^2 M}{\partial z^2}$ |
| Medium response | $P = n_0\big(\rho_{12}\,d + \rho_{21}\,d^*\big)$ | $M = n_0\big(\rho_{12}\,m + \rho_{21}\,m^*\big)$ |
| Field envelope | $E = \frac{1}{2}A\,e^{i(kz-\omega t)} + \text{c.c.}$ | $B = \frac{1}{2}A\,e^{i(kz-\omega t)} + \text{c.c.}$ |
| *Scaling quantities* | | |
| Cooperation time | $T_R = \dfrac{2\epsilon_0 c\hbar}{\omega\,\lvert d\rvert^2\, n_0\, L}$ | $T_R = \dfrac{2c\hbar}{\omega\mu_0\,\lvert m\rvert^2\, n_0\, L}$ |
| Field amplitude scale | $A_0 = -i\dfrac{\hbar}{d\,T_0}$ | $A_0 = -i\dfrac{\hbar}{m\,T_0}$ |
| Field scale | $E_0 = \dfrac{\hbar}{\lvert d\rvert\, T_0}$ | $B_0 = \dfrac{\hbar}{\lvert m\rvert\, T_0}$ |
| Medium response scale | $P_0 = n_0\,\lvert d\rvert$ | $M_0 = n_0\,\lvert m\rvert$ |
| Intensity scale | $I_0 = \dfrac{\epsilon_0 c E_0^2}{2}$ | $I_0 = \dfrac{cB_0^2}{2\mu_0}$ |
| *Physical observables (real case)* | | |
| Physical field | $E_0\,A'\sin(kz-\omega t)$ | $B_0\,A'\sin(kz-\omega t)$ |
| Medium response | $P_0\,R\cos(kz-\omega t)$ | $M_0\,R\cos(kz-\omega t)$ |
| Intensity | $I_0\,\lvert A'\rvert^2\sin^2(kz-\omega t)$ | |

In both cases the cooperation time can be expressed in terms of the single-atom spontaneous emission rate $\Gamma$. Using the standard relations

$$\Gamma = \frac{\omega^3 |d|^2}{3\pi\epsilon_0 \hbar c^3} \quad \text{(electric)}, \qquad \Gamma = \frac{\omega^3 |m|^2}{3\pi\epsilon_0 \hbar c^5} \quad \text{(magnetic)}, \tag{56}$$

substituting into the respective expressions for $T_R$ yields the same result in both cases:

$$\boxed{T_R = \frac{2\omega^2}{3\pi c^2 \, n_0 \, L \, \Gamma}.} \tag{57}$$

The cooperation time is therefore inversely proportional to $n_0$, $L$, and $\Gamma$. For a sufficiently dense and inverted medium, $T_R \ll T_1$ and the system enters the superradiant regime. In the application, $\nu$, $\Gamma$, $n_0$, and $L$ are entered in the **Sample** tab, from which $T_R$ is computed internally.

**Note**

Because the dimensionless equations (47)–(49) depend only on $\eta$, $T_1'$, $T_2'$, $w_0$, $R_0$, $\Lambda_n'$, and the boundary field $A'(0, \tau')$, the same solver describes both electric-dipole and magnetic-dipole physics.